Task 1

CJD

2025-12-06

Librarys:

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
```

Set WD and import data

```r
gdp_per_capita_df <- read.csv("Exported_data/gdp_per_capita_cleaned.csv")
gdp_per_capita_df <- gdp_per_capita_df %>%
  filter(Year >= 2015)
```

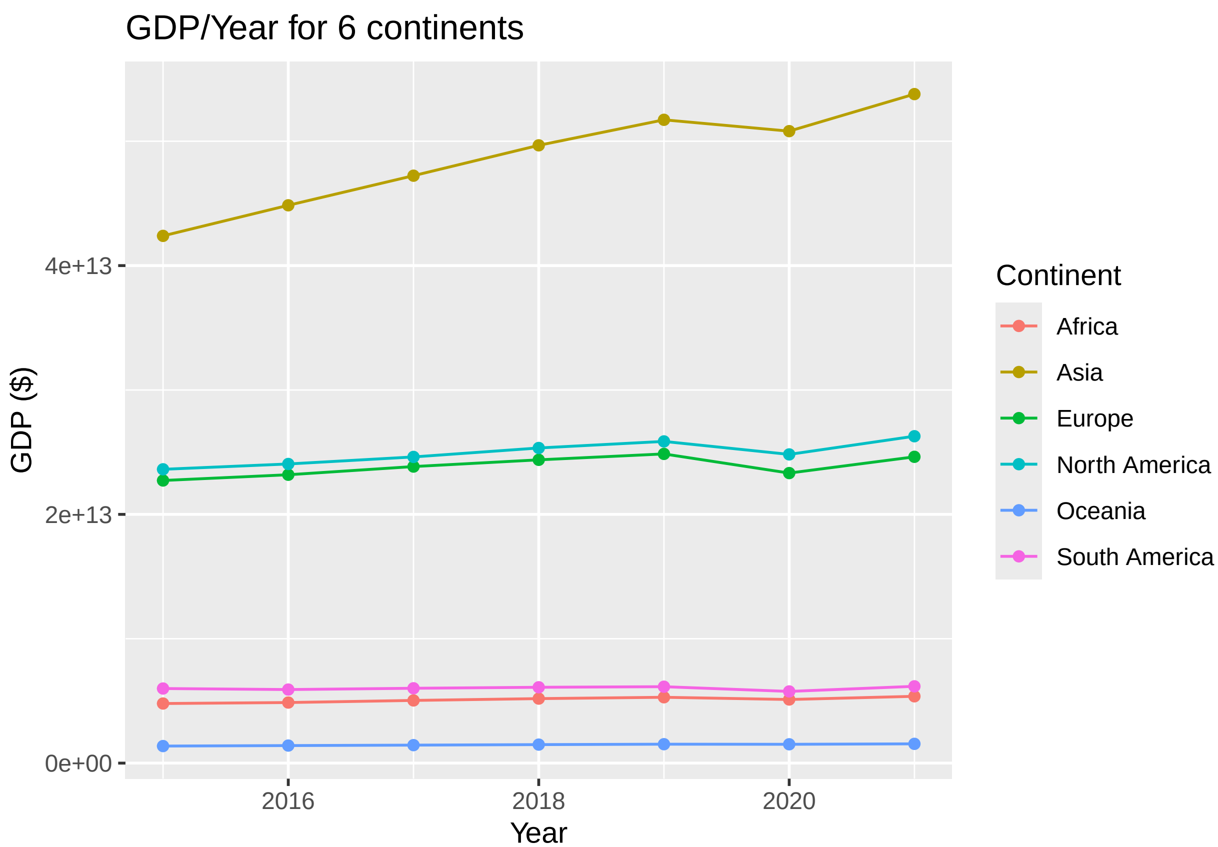
Calculate the total GDP of each continent

```r
continent_gdp_totals <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Total_GDP = sum(Gdp, na.rm = TRUE)
  ) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.
```

Plot a graph of GDP for each continent, vs time

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = Total_GDP, colour = Continent)) +
  geom_point() +
  geom_line() +
  labs(title = "GDP/Year for 6 continents", y = "GDP ($)")
```

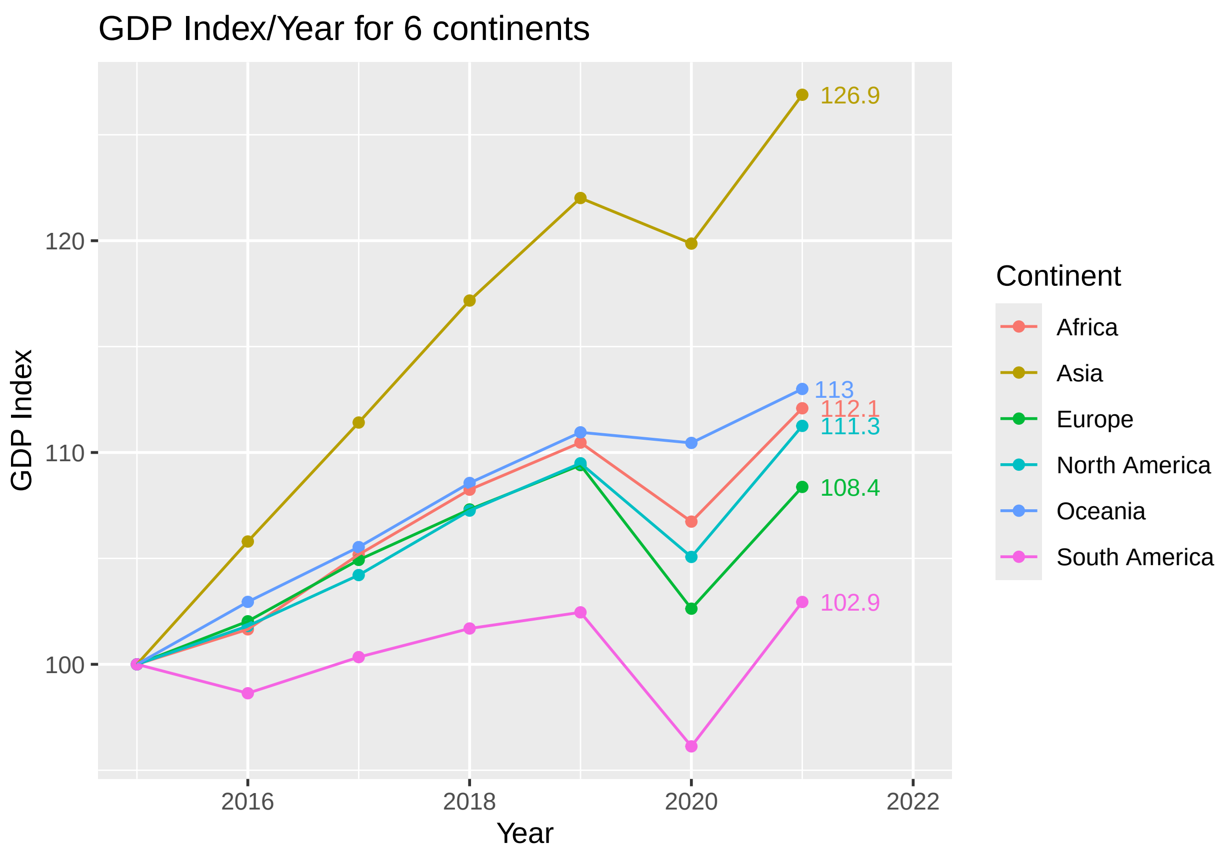


I decided this graph could show more if they all started at an index of 100

```
continent_gdp_totals <- continent_gdp_totals %>%
  group_by(Continent) %>%
  mutate(
    GDP_Index = Total_GDP / first(Total_GDP) * 100
  )
```

We now replot the graph, using normalised values

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = GDP_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_text(
  data = continent_gdp_totals %>% filter(Year == 2021),
  aes(label = round(GDP_Index, 1)),
  hjust = -0.3,
  size = 3,
  show.legend = FALSE) +
  expand_limits(x = 2022) +
  labs(title = "GDP Index/Year for 6 continents", y = "GDP Index")
```



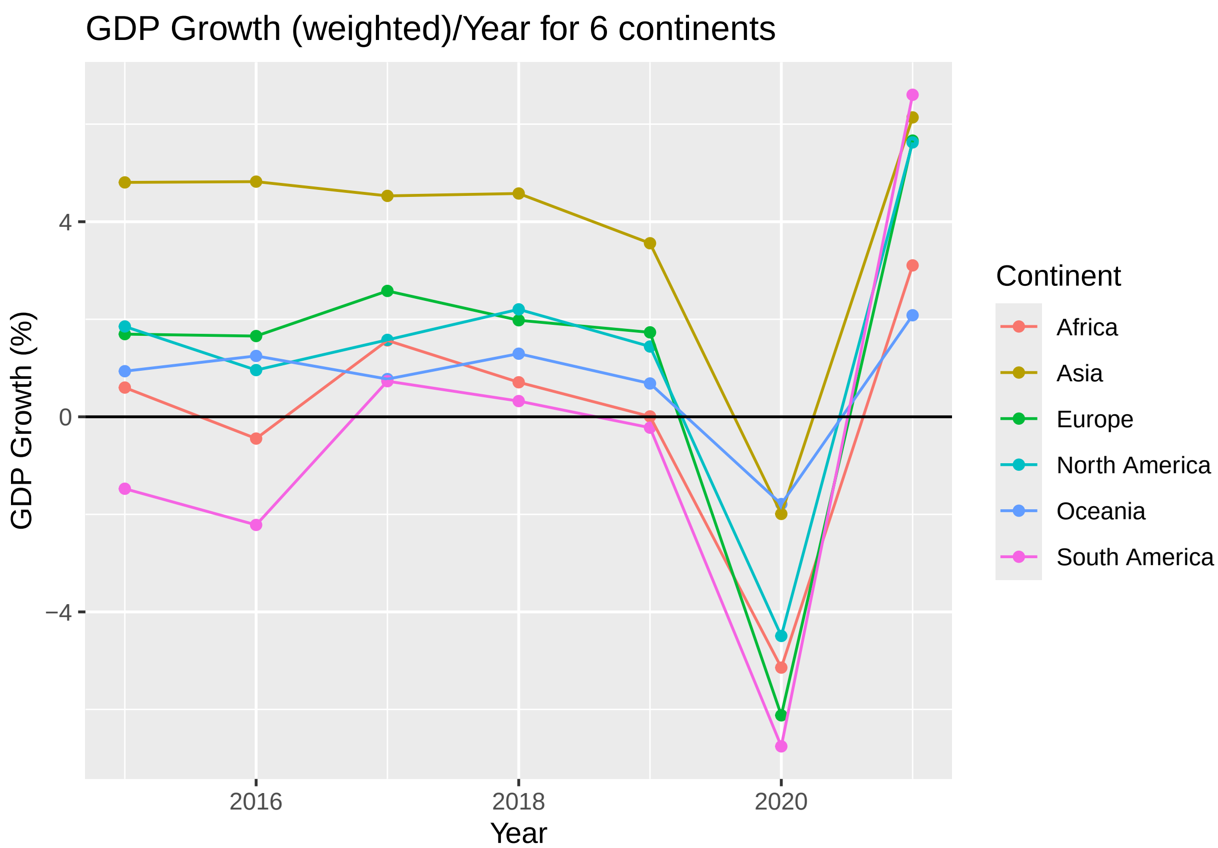
Calculate the weighted and unweighted growth of each continent

```
continents_growth_df <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Weighted_Growth = sum(Growth * Gdp, na.rm = TRUE) / sum(Gdp, na.rm = TRUE),
    Avg_Growth = mean(Growth, na.rm = TRUE)
  ) %>%
  ungroup()
```

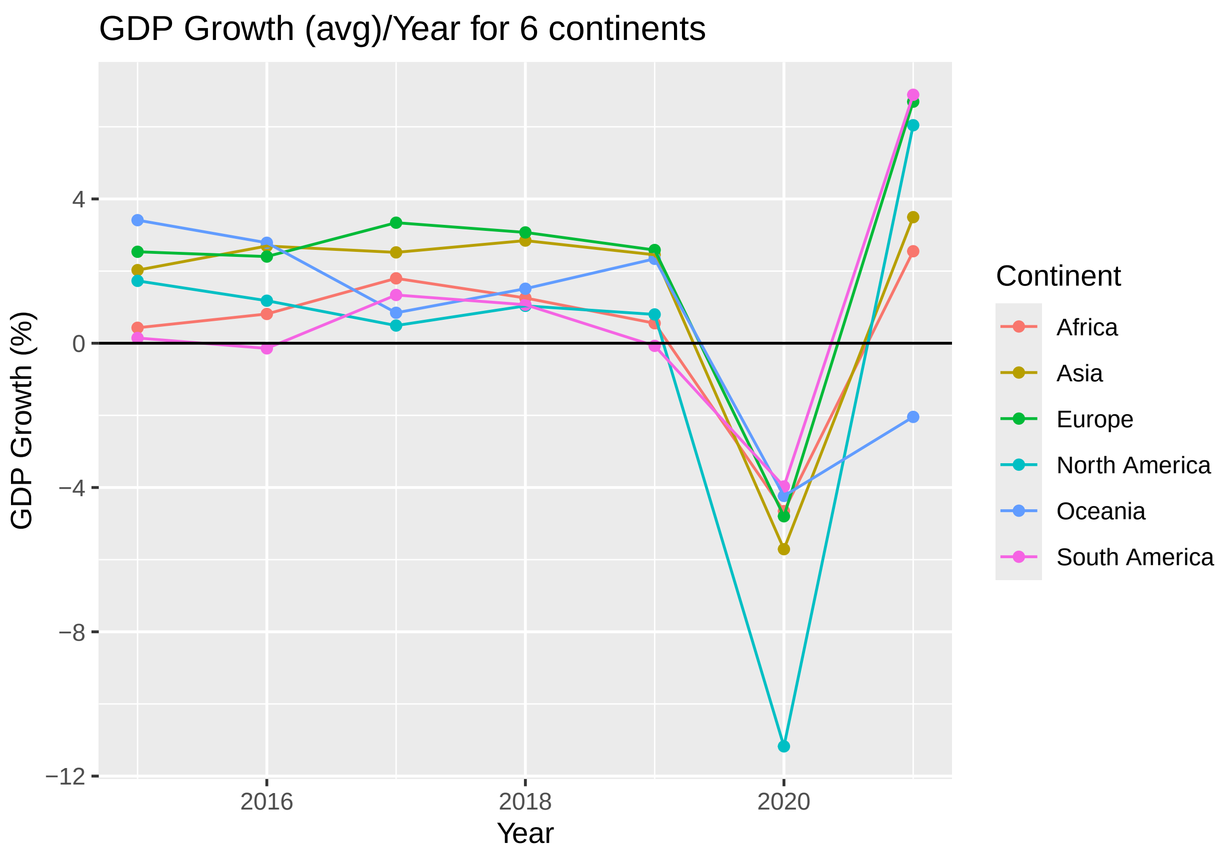
```
## 'summarise()' has grouped output by 'Continent'. You can override using the
## '.groups' argument.
```

Plot the growth graphs of each continent

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```




```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (avg)/Year for 6 continents", y = "GDP Growth (%)")
```

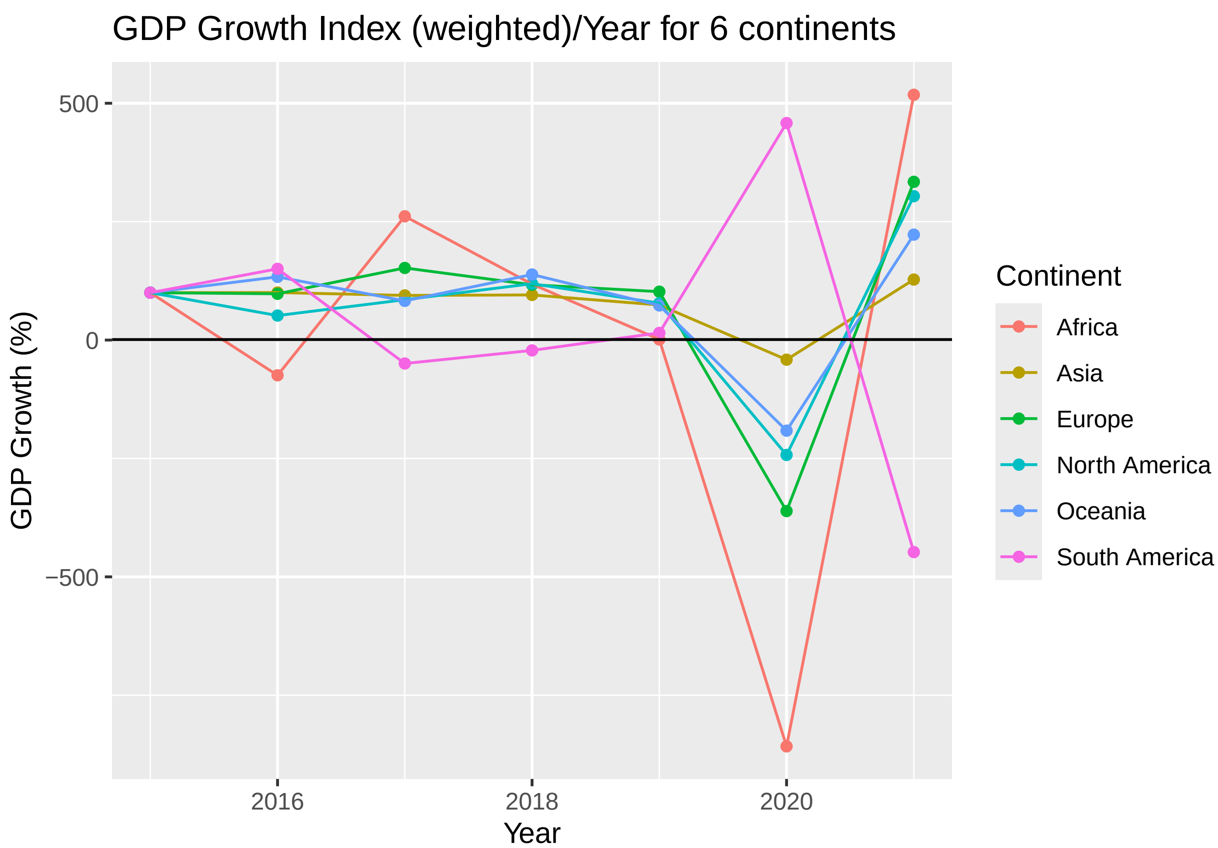


I decided to do a similar thing with growth about index (realised is redundant, don't use in analysis)

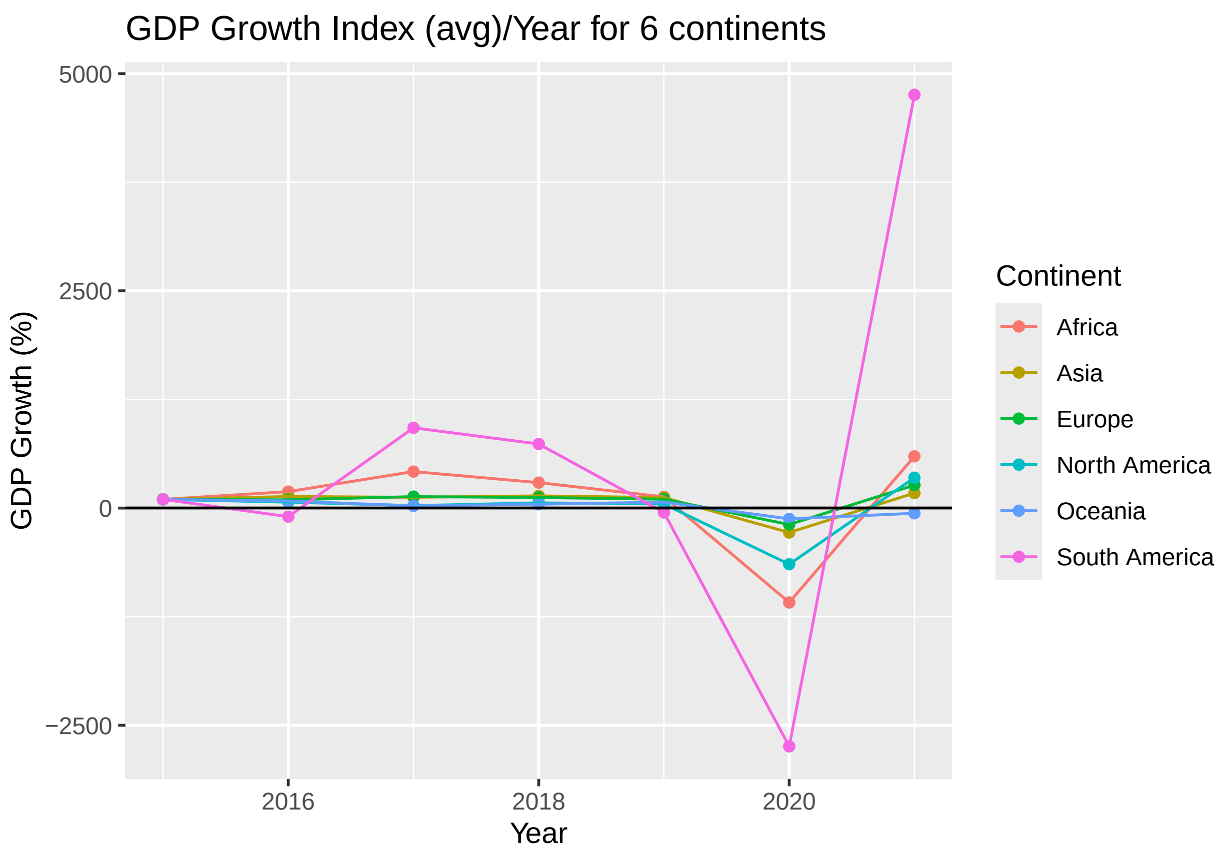
```
continents_growth_df <- continents_growth_df %>%
  group_by(Continent) %>%
  mutate(
    Avg_Growth_Index = Avg_Growth / first(Avg_Growth) * 100,
    Weighted_Growth_Index = Weighted_Growth / first(Weighted_Growth) * 100
  )
```

Plotting the graphs

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```




```r
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (avg)/Year for 6 continents", y = "GDP Growth (%)")
```



REF:

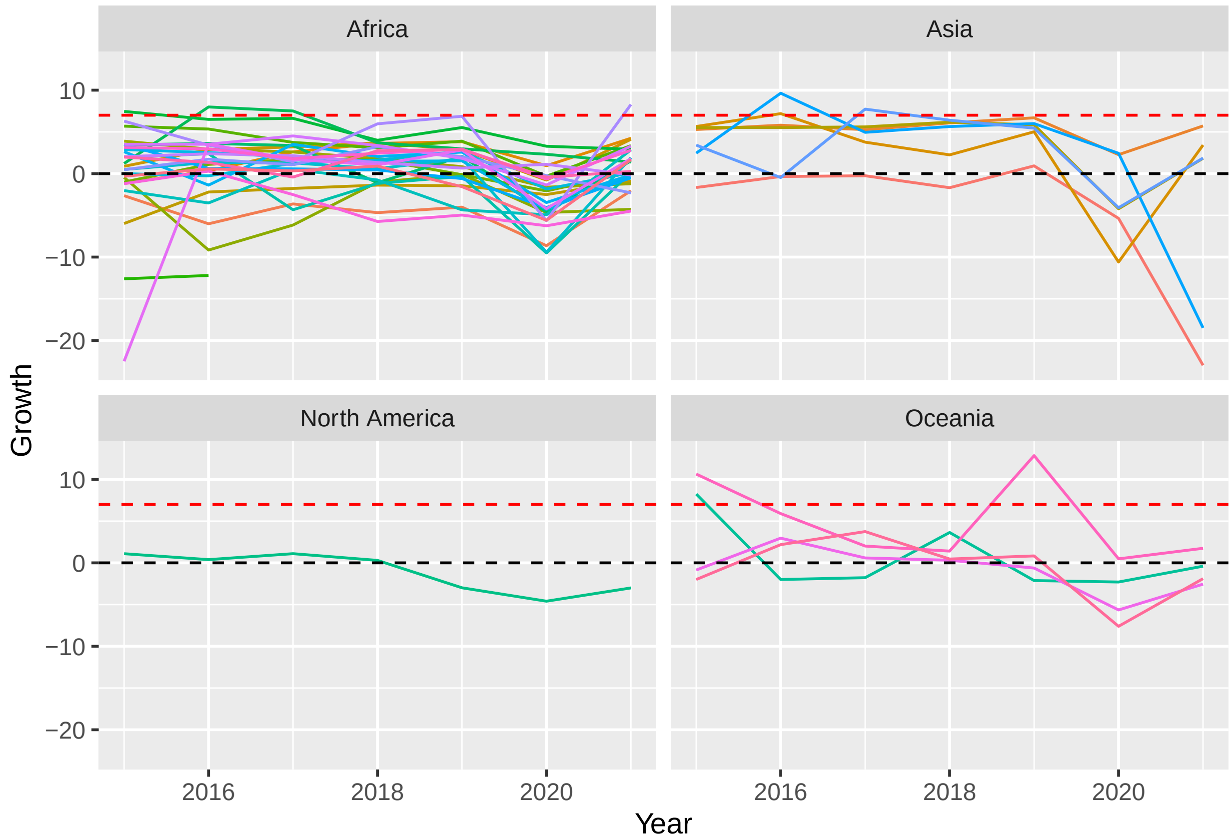
28/11/2025 - asked ChatGPT - "i have text near the last points on the x axis, how can i make it so it doesn't overlap the axis"

Plot ldc growth, each separate country in a certain continent (6 graphs) Create a ldc database

```
ldc <- gdp_per_capita_df %>%
  filter (Ldc == TRUE)
```

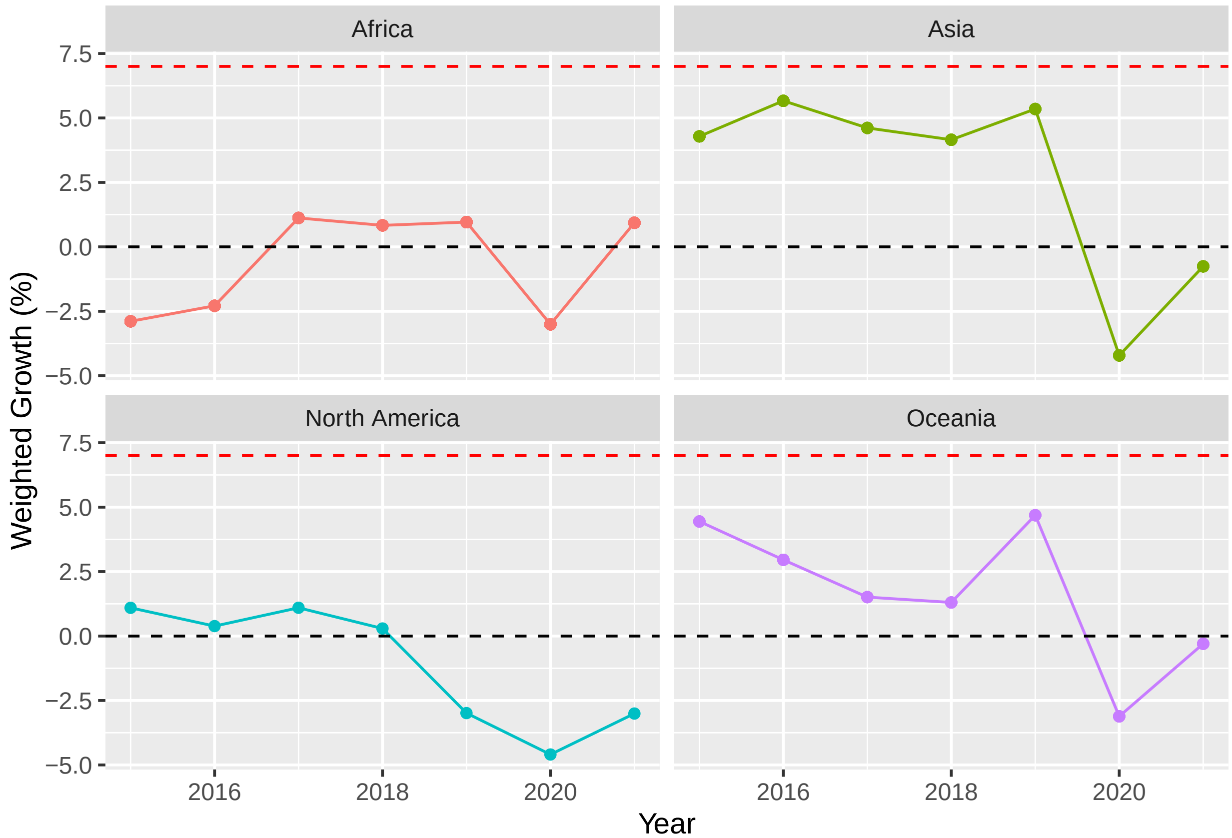
Plot the graphs

```
ggplot(ldc, aes(x = Year, y = Growth, color = Entity)) +
  geom_line() +
  facet_wrap(~ Continent) +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme(legend.position = "none")
```

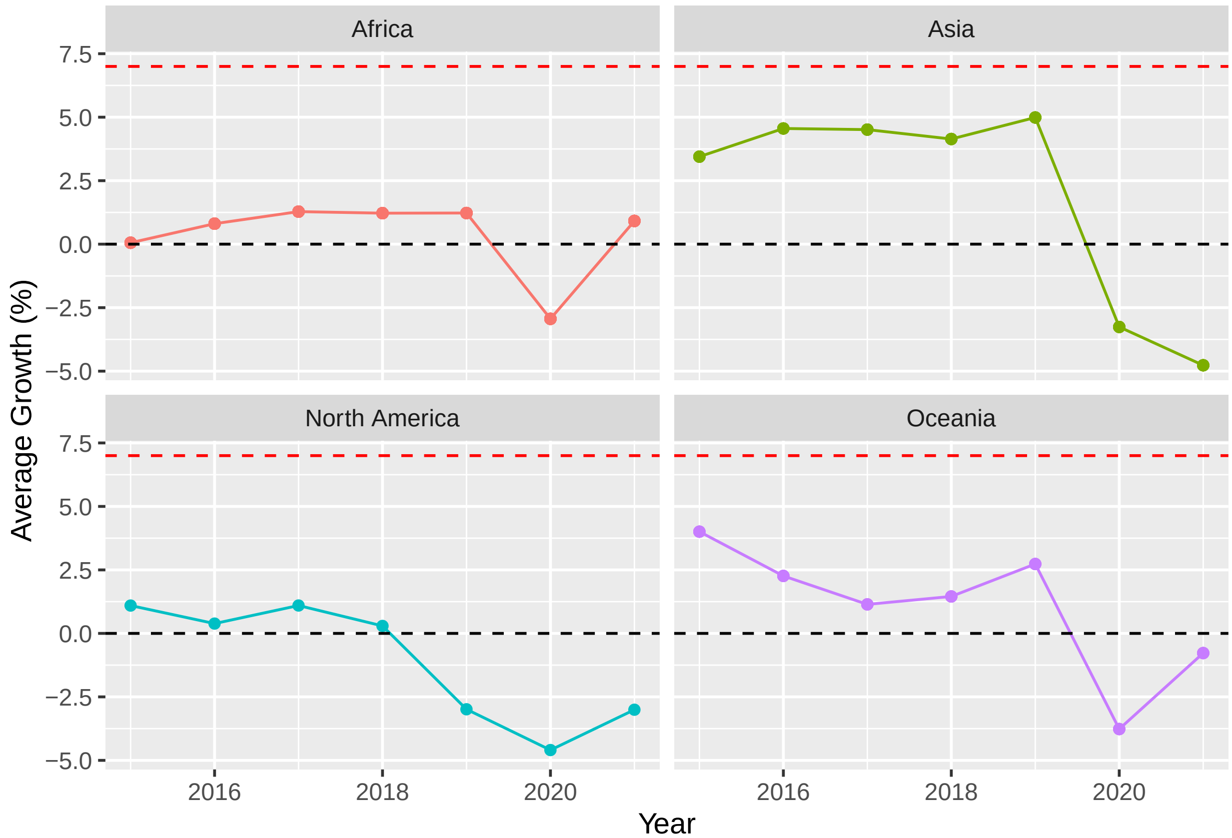


Do a normal and a weighted average of ldc growth rate for each continent, then plot a graph of each continent and how they compare

```
ldc_weighted <- ldc %>%
  group_by(Year, Continent) %>%
  mutate(
    weighted_growth = sum(Gdp_per_capita * Growth, na.rm = TRUE) / sum(Gdp_per_capita, na.rm = TRUE)
  ) %>%
  ungroup()

ggplot(ldc_weighted, aes(x = Year, y = weighted_growth, colour = Continent)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Continent) +
  theme(legend.position = "none") +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(y = "Weighted Growth (%)")
```

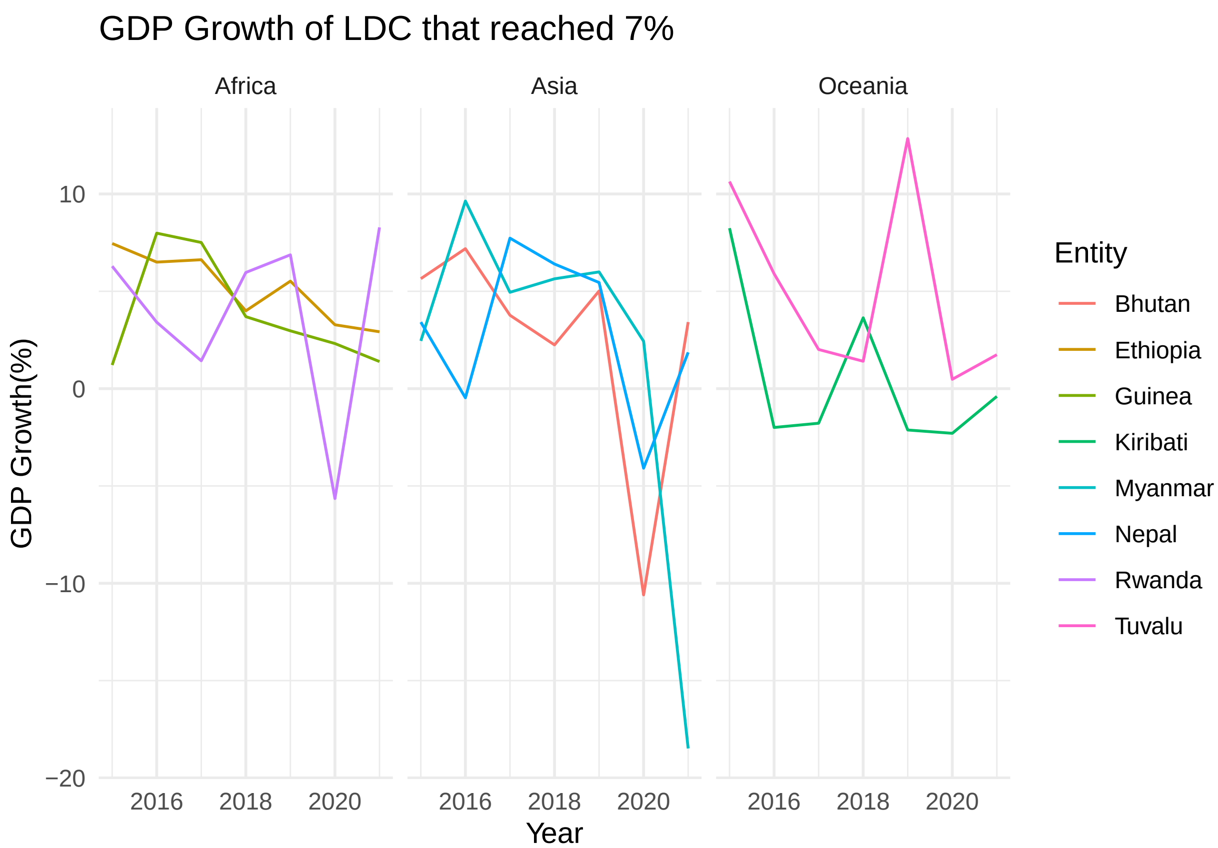



```r
ldc_avg_growth <- ldc %>%
  group_by(Year, Continent) %>%
  mutate(
    Avg_growth = mean(Growth, na.rm = TRUE)
  ) %>%
  ungroup()

ggplot(ldc_avg_growth, aes(x = Year, y = Avg_growth, colour = Continent)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Continent) +
  theme(legend.position = "none") +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(y = "Average Growth (%)")
```



Identify which LDCs have had a growth that reached 7% and plot their growth

```
ldc_7 <- ldc %>%
  group_by(Entity) %>%
  filter(any(Growth >= 7)) %>%
  ungroup()

ggplot (ldc_7, aes(x = Year, y = Growth, color = Entity)) +
          geom_line() +
          facet_wrap (~ Continent) +
          theme_minimal() +
          labs(
            title = "GDP Growth of LDC that reached 7%",
            x = "Year",
            y = "GDP Growth(%)"
          )
```



REF:

28/11/2025 - asked ChatGPT - ""I have a facet plot in ggplot with some continents having tons of lines and some only one. How can I adjust the y-axis scales so each graph uses the right space for its lines and they don't look squished or empty?"

Find the 3 countries with most GDP growth, and 3 countries with most GDP decline

```
# Top 3 growth
top_growth <- gdp_per_capita_df %>%
  group_by(Entity) %>%
  summarise(Avg_Growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(desc(Avg_Growth)) %>%
  slice(1:3)

# Top 3 decline
top_decline <- gdp_per_capita_df %>%
  group_by(Entity) %>%
  summarise(Avg_Growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(Avg_Growth) %>%
  slice(1:3)
```

Find the countries with biggest gdp in each continent

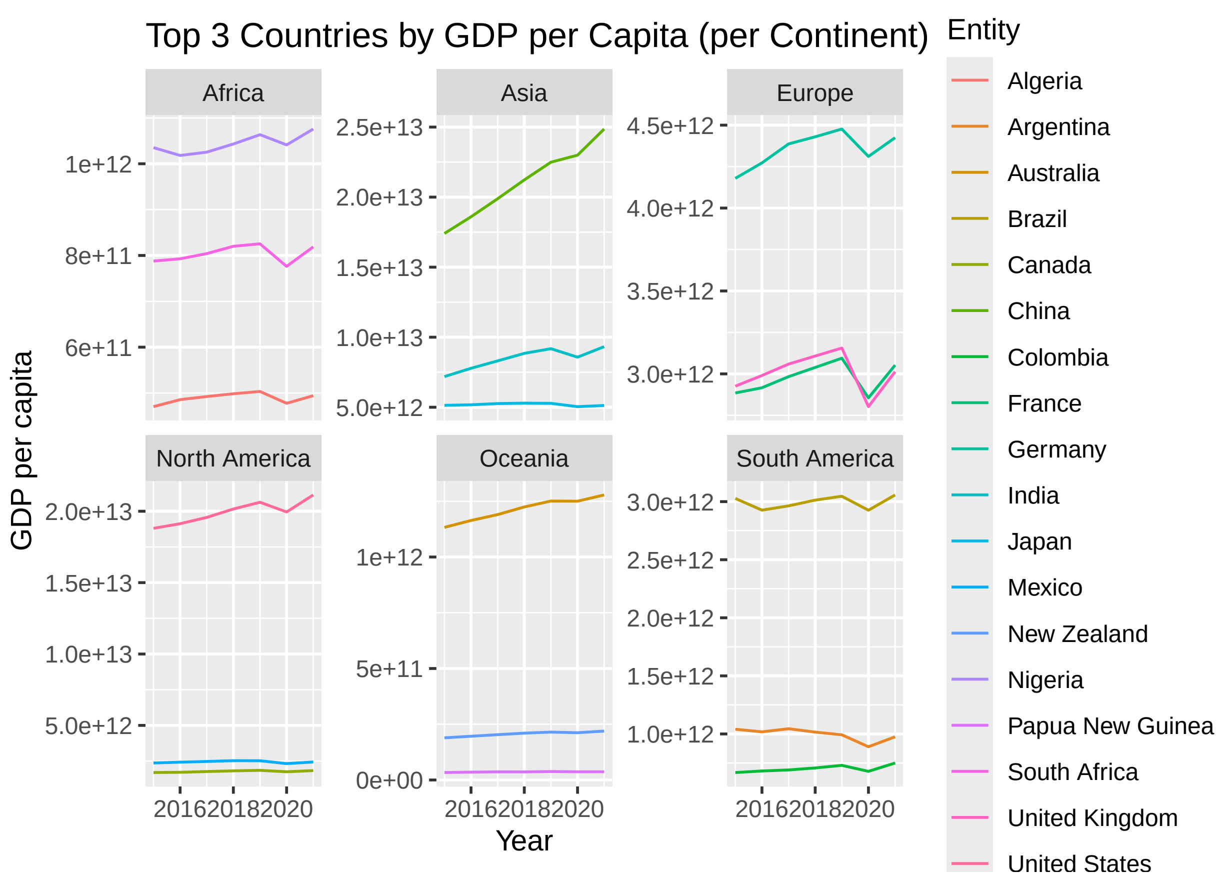
```
top_3_per_continent <- gdp_per_capita_df %>%
  group_by(Continent, Entity) %>%
  summarise(Total_GDP = sum(Gdp, na.rm = TRUE)) %>%
```

```
  arrange(Continent, desc(Total_GDP)) %>%
  group_by(Continent) %>%
  slice(1:3)
```

```
## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.
```

Plot their graphs

```
filtered_df <- gdp_per_capita_df %>%
  semi_join(top_3_per_continent, by = c("Continent", "Entity"))

ggplot(filtered_df, aes(x = Year, y = Gdp, colour = Entity)) +
  geom_line() +
  facet_wrap(~ Continent, scales = "free_y") +
  labs(title = "Top 3 Countries by GDP per Capita (per Continent)", y = "GDP per capita")
```




```
#free y scaling - allows the y-axis of a plot to vary independently for each row of facets
```

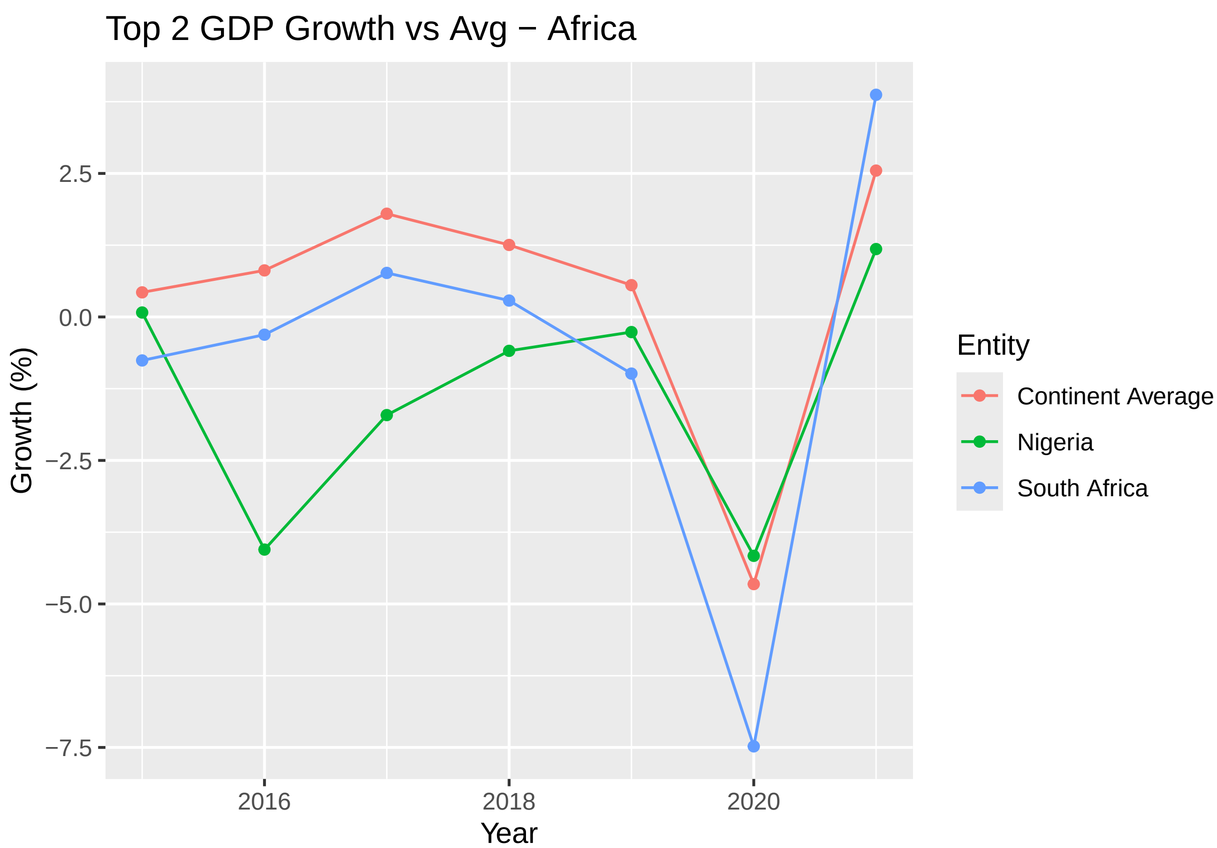
Graph to see if china and india are the drivers of asias growth

```
top_2_per_continent <- gdp_per_capita_df %>%
  group_by(Continent, Entity) %>%
  summarise(Total_GDP = sum(Gdp, na.rm = TRUE)) %>%
```

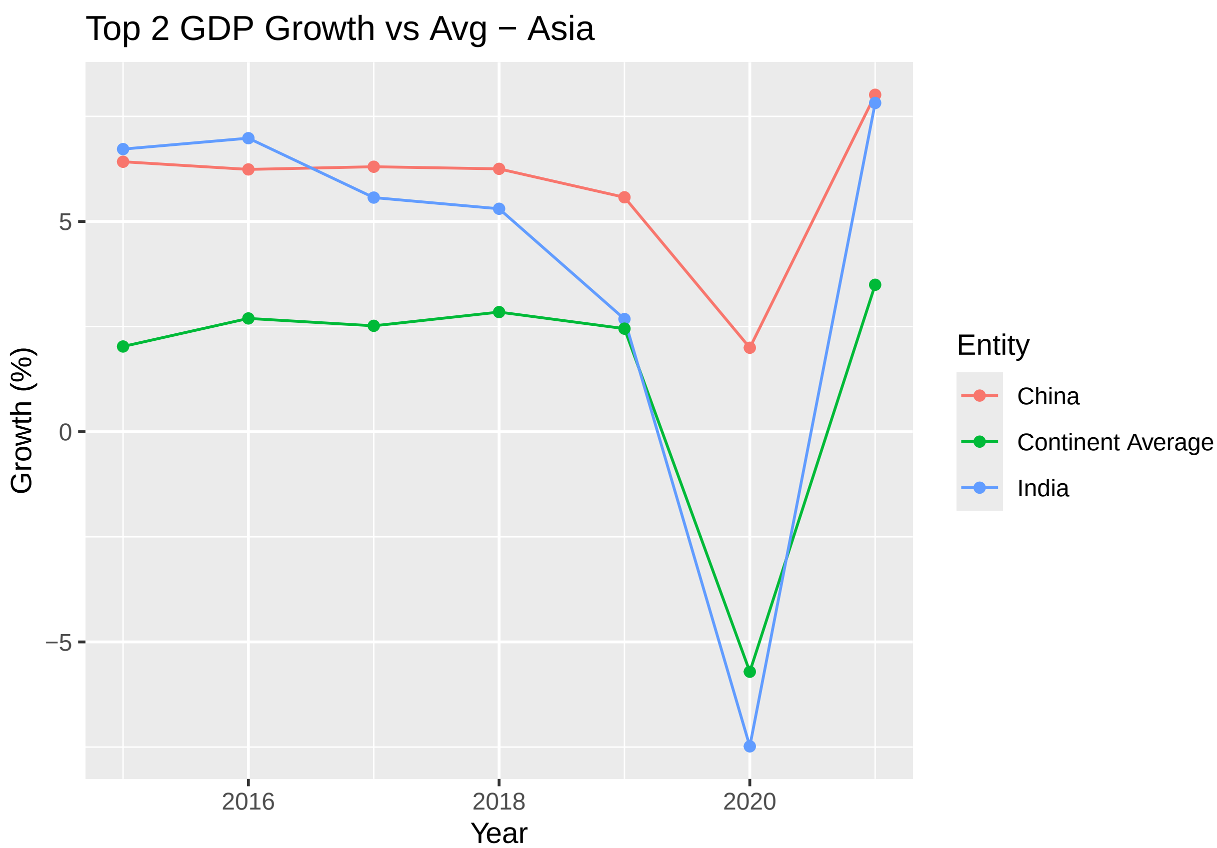
```r
  arrange(Continent, desc(Total_GDP)) %>%
  group_by(Continent) %>%
  slice(1:2) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.
```

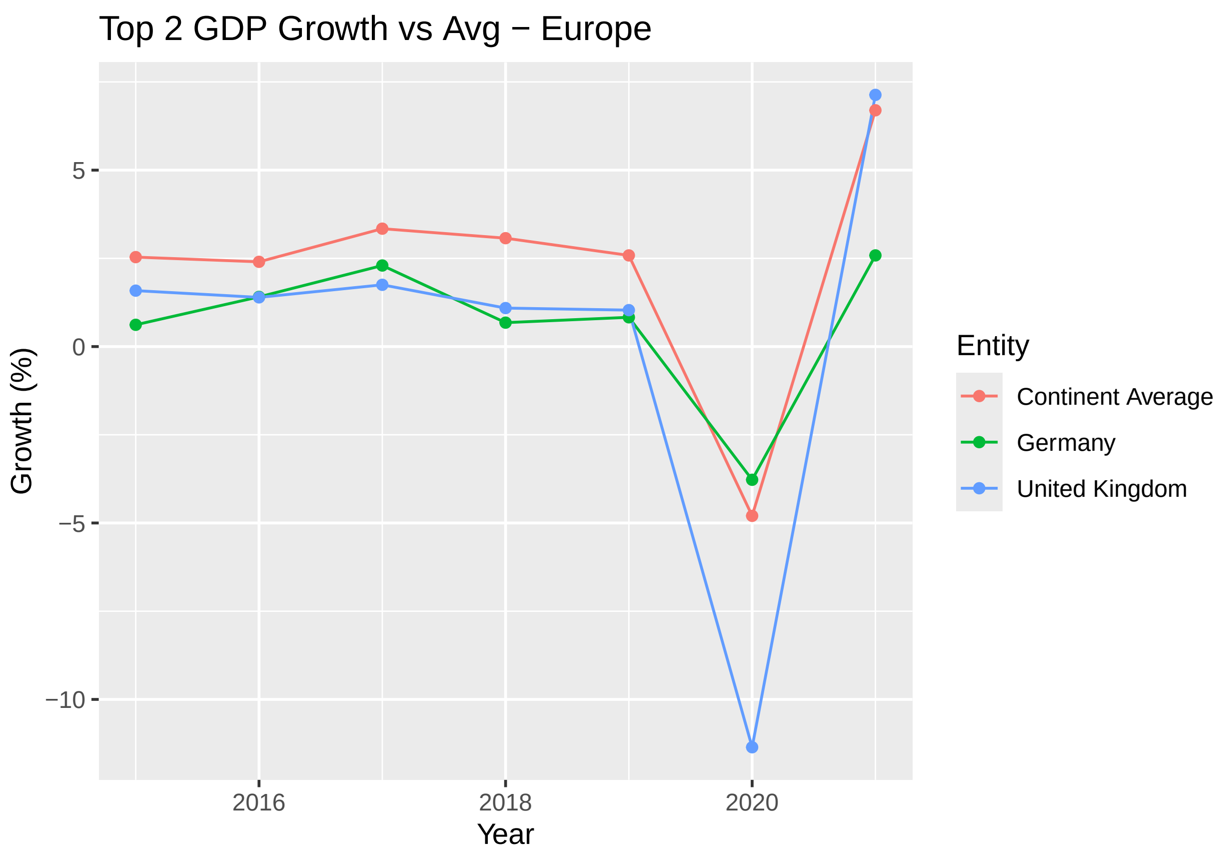
```r
get_plot_data <- function(x) {

  top_2_entities <- top_2_per_continent %>%
    filter(Continent == x) %>%
    pull(Entity)

  country_data <- gdp_per_capita_df %>%
    filter(Continent == x, Entity %in% top_2_entities)

  continent_avg <- continents_growth_df %>%
    filter(Continent == x) %>%
    rename(Growth = Avg_Growth) %>%
    mutate(Entity = "Continent Average")

  plot_data <- bind_rows(country_data, continent_avg)

  return(plot_data)
}

plot_drivers <- function(continent) {
  data <- get_plot_data(continent)

  ggplot(data, aes(x = Year, y = Growth, colour = Entity)) +
    geom_line() +
    geom_point() +
    labs(
      title = paste("Top 2 GDP Growth vs Avg -", continent),
      y = "Growth (%)"
    )
}

plot_drivers("Africa")
```



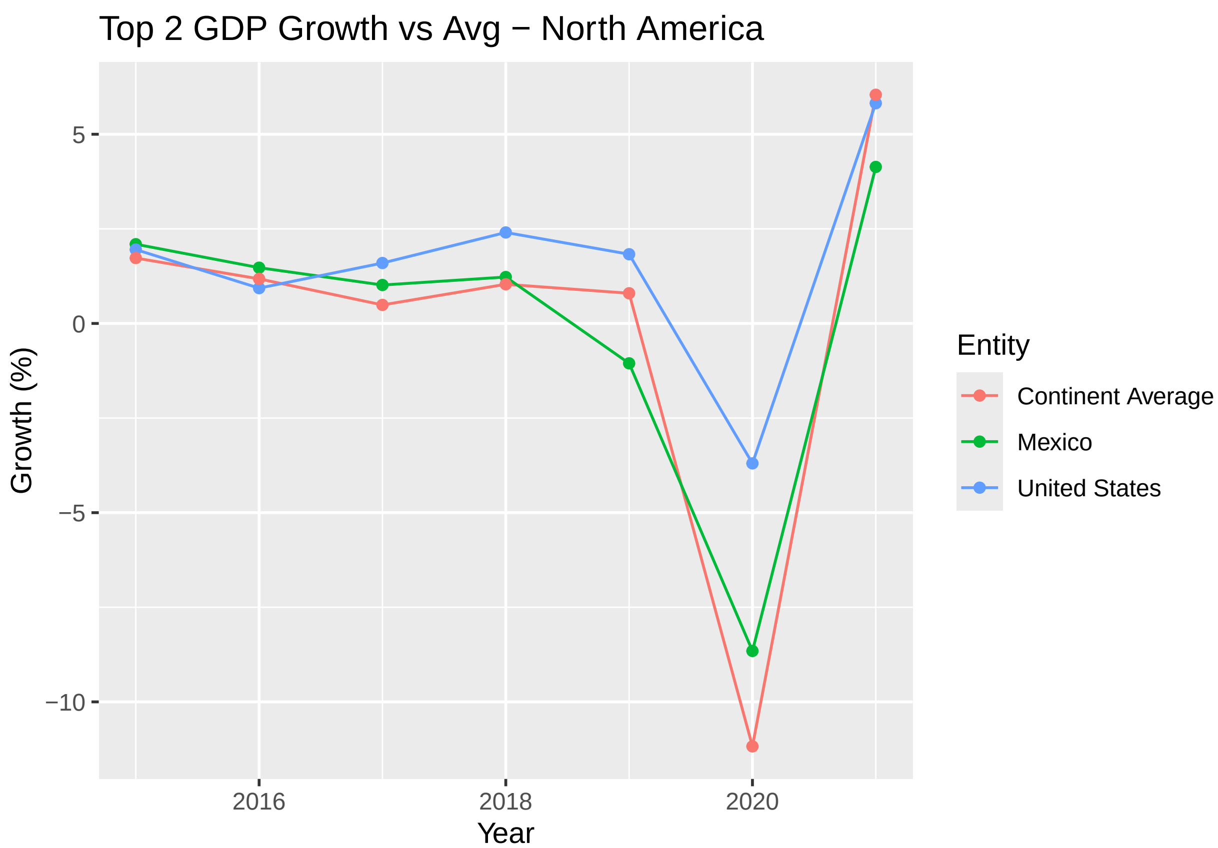

```
plot_drivers("Asia")
```



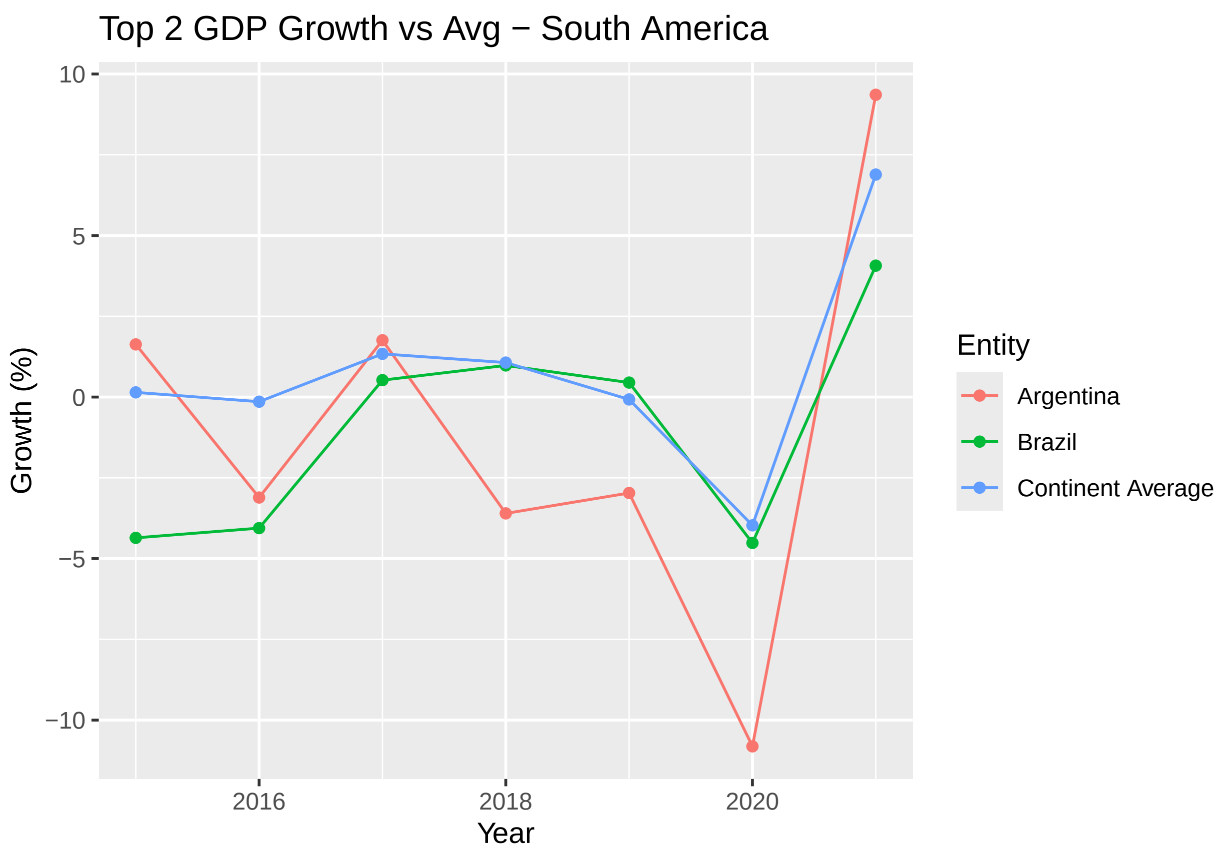

```
plot_drivers("Europe")
```



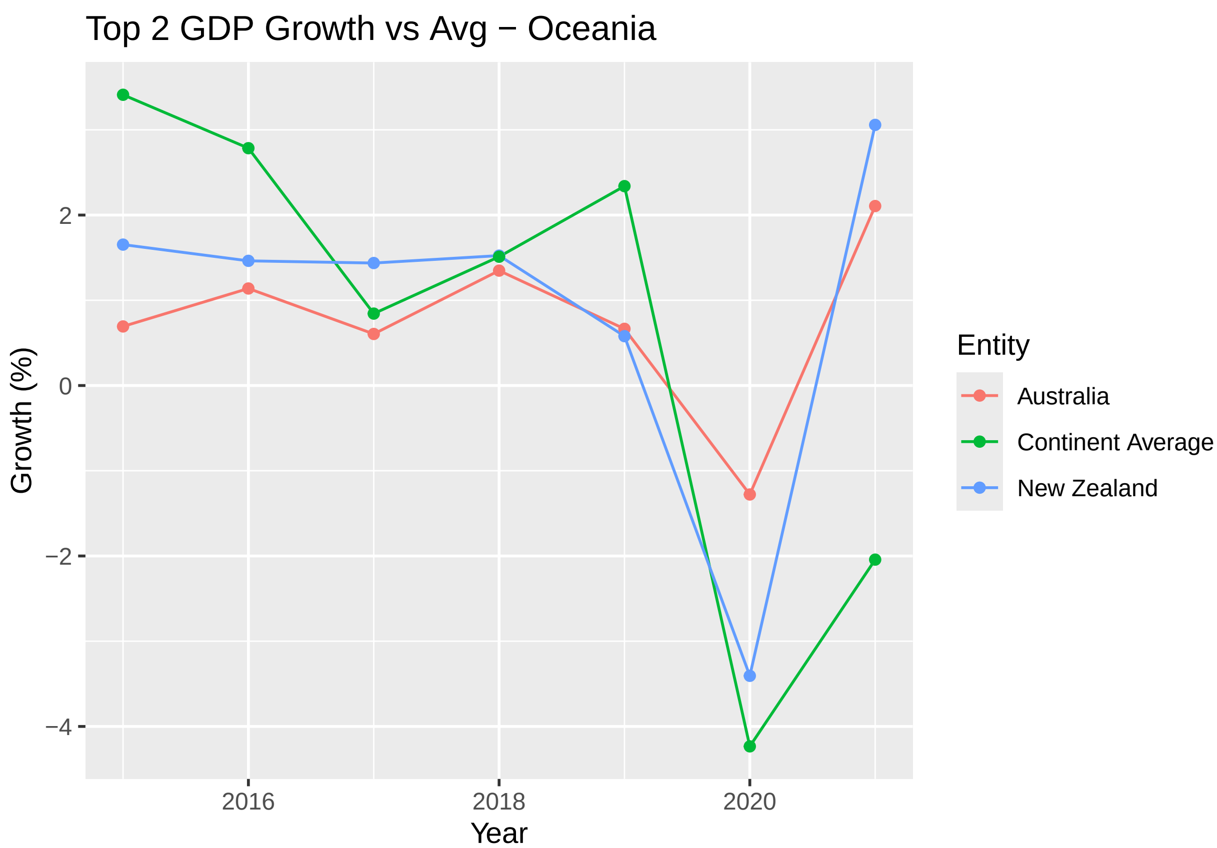

```
plot_drivers("North America")
```




```
plot_drivers("South America")
```




```r
plot_drivers("Oceania")
```



ANALYSIS Africa: Nigeria, South Africa, Algeria Asia: China, India, Japan Europe: Germany, UK, France. Germany leading North America: US, Mexico, Canada Oceania: Australia, New Zealand, Papa New Guinea South America: Brazil, Argentina, Colombia. Brazil drives volatility, steady increase

in all continents apart from Africa, there is one clear country that is driving GDP why this is important to analyse? these countries are the ones that drive continent verages the weighted averages are heavily influenced by large economics (eg US, China, Germany) spikes/dips in major economies explain most fluctations in continent graphs. this can definitely be seen in europe in thr last few years for example